

Risk-Adaptive Multi-Factor Bio-Crypt Lock

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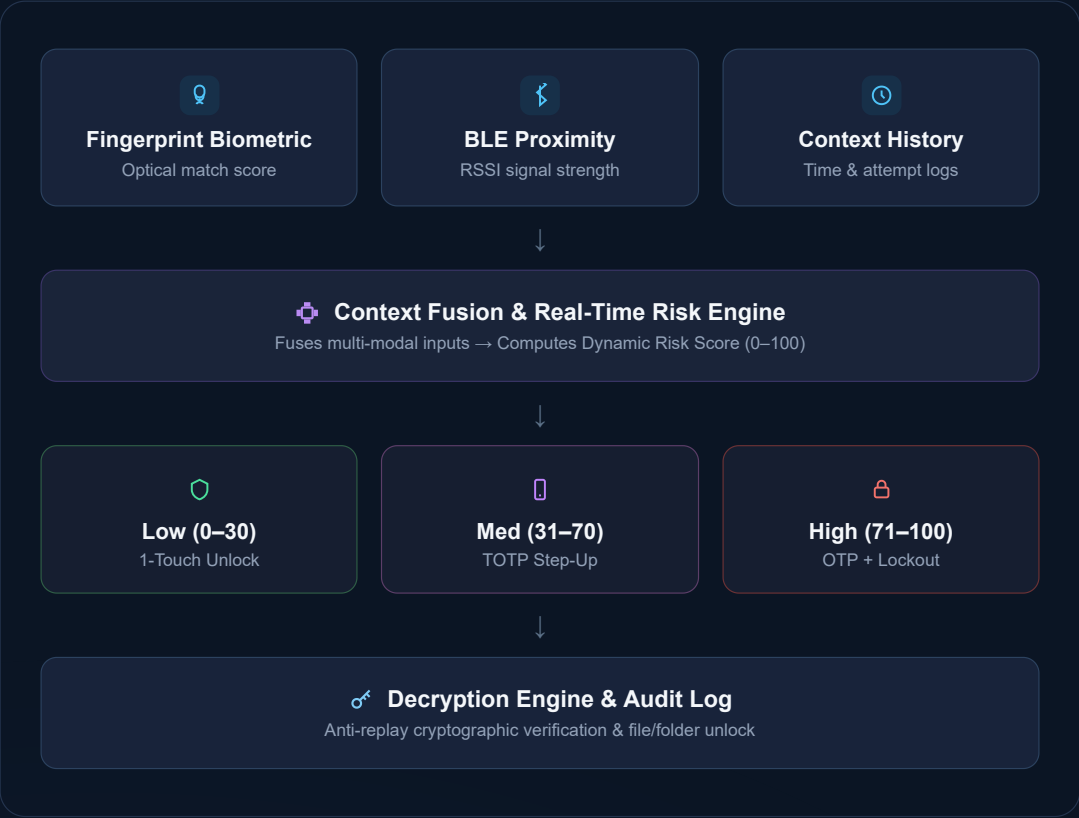
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 Guide: Professor Rathna R

Bio-Crypt Lock: Core Architectural Idea

Concept Architecture



Multi-Factor Context Fusion

Combines fingerprint biometric recognition with Bluetooth proximity status and behavioral history into a unified evaluation.



Real-Time Risk Engine

Calculates dynamic risk scores (0-100) to automatically adjust authentication strictness based on environmental context.



3-Tier Risk-Adaptive Auth

Low (0-30): Seamless single-touch unlock — no OTP required.
Medium (31-70): Dynamic OTP delivered to the trusted paired device for verification.
High (71-100): Strong OTP challenge delivered to the trusted device + lockout protocol.



Access-Control Software Integration

Deploys risk-adaptive authentication logic directly into the access-control software with anti-replay cryptographic defenses.

Adaptive Risk-Based Multi-Layer Authentication Framework for Secure Online Banking

Soyemi, Hammed & Soyemi (2026) — *FUDMA Journal of Sciences*, 10(4), 331–338.

[Open Paper Link](#)

doi.org/10.33003/fjs-2026-1004-4...

✓ Positive Features

- Validates variable authentication strictness with empirical performance (98.9% accuracy, EER 1.65%, AUC 0.991).
- Utilizes a probabilistic model to calculate attack-success likelihood per authentication layer.
- Proves that dynamic risk-adaptive authentication significantly outperforms static single- and two-factor baselines.

✗ Negative Features

- Designed exclusively for web/online banking software, ignoring IoT resource constraints and hardware locks.
- Relies on PIN and graphical passwords rather than biometrics, omitting fingerprint spoofing and sensor noise risks.
- Excludes Bluetooth proximity, credential replay defenses, and non-invertible biometric template security.

✿ Research Gap Relative to Bio-Crypt Lock

Proves dynamic risk engines reduce online fraud, but fails to extend this concept to a software-based file/folder access-control system. Your project fills this gap by deploying risk-adaptive logic onto a file/folder unlock system where the primary factor is a fingerprint and the context signal includes Bluetooth proximity.

Literature Review (2/5)

Mobile OTP + Fingerprint

Multi-Factor Authentication Architecture Design with OTP and Fingerprint Integration

ICoBITS (2025) — *Proceedings of ICoBITS 2025 / ResearchGate.*

[Open Paper Link](#)

icobits.ubhinus.ac.id/.../348

✓ Positive Features

- Confirms fingerprint biometric + OTP as an effective, highly secure two-factor combination.
- Designed specifically for mobile applications with low computational overhead and high scalability.
- References dynamic risk profiling as a foundation to justify hybrid multi-factor authentication models.

✗ Negative Features

- Uses a static OTP length that does not adapt in complexity according to a real-time risk score.
- Targeted strictly at smartphone mobile app login rather than physical smart lock hardware and Bluetooth sensing.
- Omits anti-replay protection, non-invertible biometric storage, and access logging features.

✚ Research Gap Relative to Bio-Crypt Lock

Stops at static two-factor pairing (fingerprint + static OTP). Your project fills this gap by dynamically scaling OTP length and complexity (6-digit vs 8-digit vs additional challenge) based on real-time risk scores fused with Bluetooth proximity context.

Literature Review (3/5)

Adaptive Auth Survey

SoK: Systematic Review of Context & Behaviour-Aware Adaptive Authentication

Podapati, Nigam & Das (2025/2026) — *HA/SA 2025*, Springer IFIP AICT vol. 761 / arXiv:2507.21101.

[Open Paper Link](#)

arxiv.org/abs/2507.21101

✓ Positive Features

- Formally grounds device proximity as a key contextual signal for adaptive authentication step-up/step-down.
- Provides a comprehensive synthesis of 41 peer-reviewed studies across 7 analytical security dimensions.
- Highlights machine learning usage (64.3% of studies) for continuous user verification and risk assessment.

✗ Negative Features

- Systematization of Knowledge (survey) paper only; contains no hardware prototype or implementation.
- Explicitly notes existing adaptive auth research is fragmented and rarely translated to deployable real-world systems.
- Focuses on smartphone OS environments without addressing smart lock hardware, OTP protocols, or biometric storage.

✿ Research Gap Relative to Bio-Crypt Lock

Highlights that adaptive authentication research remains theoretical and lacks concrete deployments. Your project directly addresses this gap by converting abstract context-aware proximity concepts into a working, deployable Bio-Crypt Lock system for securing files and folders.

Literature Review (4/5)

BLE Smart Door Lock

Smart Door Lock Innovation Using Integration of BLE and MQTT IoT Protocol

Kurniasyah & Rakhmawati (2025) — *Indonesian Journal of Electrical & Electronics Engineering*, 8(1), 41–46.

[Open Paper Link](#)

doi.org/10.26740/inajece.v8n1

✓ Positive Features

- Validates practical hardware viability using ESP32, fingerprint reader, and BLE proximity RSSI sensing.
- Demonstrates high proximity sensing accuracy (avg RSSI error 1.68 dBm) with empirical environmental testing.
- Includes sensor reliability metrics and network QoS benchmarking for real-time physical access control.

✗ Negative Features

- Treats BLE, fingerprint, RFID, and keypad as independent parallel unlock paths with zero risk fusion.
- Completely lacks OTP credentials, leaving RFID and keypad channels vulnerable to replay attacks.
- Does not implement non-invertible biometric template protection or structured audit logging.

✖ Research Gap Relative to Bio-Crypt Lock

Proves hardware feasibility of BLE proximity and fingerprint sensors, but fails to fuse them into a unified risk engine. Your project bridges this gap by combining fingerprint score and BLE trust into a single score that determines secondary OTP requirement.

Literature Review (5/5)

MFA Vulnerability Analysis

Excavating Vulnerabilities Lurking in Multi-Factor Authentication Protocols

Ang, Chekole & Zhou (2024) — *Singapore University of Technology & Design* / arXiv:2407.20459.

[Open Paper Link](#)

arxiv.org/abs/2407.20459

✓ Positive Features

- Establishes an 8-criteria security evaluation checklist to audit multi-factor authentication protocols.
- Recommends dynamic single-use rapid-expiry OTPs to effectively neutralize credential replay attacks.
- Warns against raw biometric template storage, advocating non-invertible cryptographic key derivation.

✗ Negative Features

- Purely analytical vulnerability analysis; provides no prototype, hardware build, or smart lock design.
- Focuses broadly on cloud/banking protocols rather than physical fingerprint-BLE embedded access locks.
- Fails to analyze latency and usability trade-offs when implementing complex cryptography on microcontrollers.

✿ Research Gap Relative to Bio-Crypt Lock

Proves why existing MFA protocols fail due to replay attacks and template leakage, but stops at theoretical recommendations. Your project operationalizes these mitigations inside a real, working file/folder access-control software system.

Project Value Proposition

Uniqueness & Objectives

💎 Core Uniqueness of the Bio-Crypt Lock Idea

Unlike existing systems that use static multi-factor checks or independent parallel unlock paths, your Bio-Crypt Lock introduces **contextual risk-based multi-modal fusion**. It combines fingerprint confidence, Bluetooth proximity trust, and behavior history into a single 0–100 risk engine that **dynamically scales OTP length (No OTP → 6-digit → 8-digit → Challenge)** while enforcing anti-replay single-use tokens delivered to the trusted paired device and non-invertible biometric storage within the access-control software.

🎯 Goal 1: Optimized Usability-Security Balance

Achieves completely frictionless, single-touch entry during low-risk scenarios (trusted Bluetooth device present + high biometric match score), while automatically stepping up authentication friction only when unusual or high-risk context anomalies are detected.

🛡️ Goal 2: Protocol & System Attack Resilience

Eliminates major access-control vulnerabilities by preventing OTP replay attacks (via 30–60s single-use tokens delivered to the trusted device) and safeguarding user biometrics against device/storage compromise (by deriving non-invertible cryptographic keys instead of storing raw templates).